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23R-D25 (MAT5304C)

2025

MATHEMATICS

(Major-III)

Paper : MAT5304C

(Numerical Analysis - I)

Full Marks : 45

Time : Two hours

***The figures in the margin indicate
full marks for the questions.***

1. Answer the following questions: $1 \times 4 = 4$
- (a) What is the order of convergence of Regular-Falsi method?
 - (b) Give one advantage and disadvantage of Bisection method.
 - (c) What is interpolation?
 - (d) If degree of the polynomial $P_n(x)$ is less than equal to n then find the value of $\Delta^{n+1}P_n(x)$.

2. Answer the following question : $2 \times 3 = 6$

(a) Under what conditions Newton-Raphson method converges?

(b) What do you mean by Round-off error in Numerical differentiation?

(c) Prove that $\Delta + \nabla = \frac{\Delta}{\nabla} + \frac{\nabla}{\Delta}$.

3. Answer the following questions : $5 \times 3 = 15$

(a) Evaluate $\frac{\Delta^2}{E} \sin(x+h) + \frac{\Delta^2 \sin(x+h)}{E \sin(x+h)}$.

(b) Discuss the conditions under which different numerical differentiation methods are used. What do you mean by error in numerical differentiation?

$3 + 2 = 5$

Or

(c) Given the table of $y = \sinh x$ with spacing $h = 0.1$, find the values of the derivatives y' and y'' at the points $x = 0.0$, $x = 0.2$ and $x = 0.5$.

x :	0.0	0.1	0.2	0.3	0.4	0.5
y :	0.00000	0.10017	0.20134	0.30452	0.41075	0.52110

- (d) Discuss order and rate of the Newton-Raphson method for approximating solution of equation.

Or

- ✓(e) Use the Newton-Raphson method to deduce the iterative procedure

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right) \text{ for evaluating } \sqrt{a} \text{ as}$$

the solution of the equation $x^2 - a = 0$.

Answer **either** (a) **or** (b) part :

4. (a) (i) Find the remainder term in Newton's forwards interpolation formula. 5

- (ii) A third degree polynomial $p(x)$ passes through the points (0,-1), (1,1), (2,1) and (3,-2). Find the polynomial. 5

- ✓(b) Find Lagrange's interpolating polynomial. Hence find the Lagrange's interpolating polynomial using following table :

$x:$	-1	0	2	5	
$f(x):$	9	5	3	15	$5+5=10$

5. (a) Find the Simpson's one-third rule. Using this to compute the value of π

form the formula $\frac{\pi}{4} = \int_0^1 \frac{dx}{1+x^2}$.

5+5=10

Or

- (b) Find the Simpson's three-eight rule. Using this to obtain an approximate

value of $\int_0^{0.3} (1-8x^3)^{\frac{1}{2}} dx$.

5+5=10